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                    SETTING UP THE ENVIRONMENT AND PREPROCESS DATA


import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
import seaborn as sns
import matplotlib.pyplot as plt
df = sns.load_dataset("titanic")
print(df.head())
print("\n========== Dataset Information ==========")
print(df.info())
print("\n========== Statistical Summary ==========")
print(df.describe())
print("\n========== Missing Values ==========")
print(df.isnull().sum())
df["age"] = df["age"].fillna(df["age"].median())
df["embark_town"] = df["embark_town"].fillna(df["embark_town"].mode()[0])
df.drop(columns=["deck"], inplace=True)
le = LabelEncoder()
df["sex"] = le.fit_transform(df["sex"])
df["embark_town"] = le.fit_transform(df["embark_town"])
df.drop(columns=["embarked", "class", "who", "alive", "adult_male", "alone"],
inplace=True)
scaler = StandardScaler()
numerical_cols = ["age", "fare"]
df[numerical_cols] = scaler.fit_transform(df[numerical_cols])
X = df.drop("survived", axis=1)
y = df["survived"]
X_train, X_test, y_train, y_test = train_test_split(
X,
y,
test_size=0.2,
random_state=42
)
print("\nTraining Data Shape:", X_train.shape)
print("Test Data Shape:", X_test.shape)
print("\nFirst 5 Rows of Training Data:")
print(X_train.head())
```

Output:

```
  survived  pclass     sex   age  sibsp  parch     fare embarked
class  \
0         0       3    male  22.0      1      0   7.2500        S
Third
1         1       1  female  38.0      1      0  71.2833        C
First
2         1       3  female  26.0      0      0   7.9250        S
Third
3         1       1  female  35.0      1      0  53.1000        S
First
4         0       3    male  35.0      0      0   8.0500        S
Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True

========== Dataset Information ==========
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
```

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 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None

========== Statistical Summary ==========
         survived      pclass         age       sibsp       parch
fare
count  891.000000  891.000000  714.000000  891.000000  891.000000
891.000000
mean     0.383838    2.308642   29.699118    0.523008    0.381594
32.204208
std      0.486592    0.836071   14.526497    1.102743    0.806057
49.693429
min      0.000000    1.000000    0.420000    0.000000    0.000000
0.000000
25%      0.000000    2.000000   20.125000    0.000000    0.000000
7.910400
50%      0.000000    3.000000   28.000000    0.000000    0.000000
14.454200
75%      1.000000    3.000000   38.000000    1.000000    0.000000
31.000000
max      1.000000    3.000000   80.000000    8.000000    6.000000
512.329200

========== Missing Values ==========
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
```

```
embark_town      2
alive            0
alone            0
dtype: int64

Training Data Shape: (712, 7)
Test Data Shape: (179, 7)

First 5 Rows of Training Data:
     pclass  sex       age  sibsp  parch      fare  embark_town
331       1    1  1.240235      0      0 -0.074583            2
733       2    1 -0.488887      0      0 -0.386671            2
382       3    1  0.202762      0      0 -0.488854            2
704       3    1 -0.258337      1      0 -0.490280            2
813       3    0 -1.795334      4      2 -0.018709            2
```